

Exploring the Impact of Virtual Reality-based Simulated Symptoms Towards Schizophrenia on Public Empathy

Jen-Chu Hsu*
National Tsing Hua University,
Hsinchu Taiwan
jchsu@gapp.nthu.edu.tw

Wernhuar Tarng
National Tsing Hua University,
Hsinchu Taiwan
whtang@mx.nthu.edu.tw

Kuo-Liang Ou
National Tsing Hua University,
Hsinchu Taiwan
klou@mx.nthu.edu.tw

ABSTRACT

Empathy is the ability to perceive and understand other people's emotions and thoughts. Virtual reality (VR), as a new technology, offers a unique opportunity to induce empathy in individuals by allowing them to immerse themselves in realistic situations from another person's perspective. In this study, we have developed a virtual reality system that incorporates situational learning and immersive interaction. Through first-person perspective simulations, participants can experience the challenges and situations encountered by individuals with positive symptoms of schizophrenia. The aim is to investigate the potential for altering public empathy and attitudes through this simulated experiential system. We hope our study could provide a new approach to health education strategies to enhance public empathy and attitudes toward people with schizophrenia.

CCS CONCEPTS

• **Human-centered computing** → Human-computer interaction (HCI); Interaction paradigms; Virtual reality.

KEYWORDS

Virtual Reality, Empathy, Schizophrenia, Positive Symptoms

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1 INTRODUCTION

According to the World Health Organization (WHO) in 2022, schizophrenia affects approximately 24 million people globally, which accounts for 1 in 300 individuals (0.32%). Although schizophrenia is not as prevalent in adults as many other mental disorders, the rate among adults is still significant at 1 in 222 (0.45%). Unfortunately, stigma and discrimination against schizophrenia persist in today's society according to the World Health Organization

[16]. The general public often lacks an understanding of schizophrenia, which indirectly affects patients' willingness to seek treatment and hampers their recovery progress. To address this issue, a group of Taiwanese public health professors, including Zhuang, Wong, Cheng and Pan [19], emphasized the need for public education programs to reduce stereotypical and discriminatory attitudes toward mental illness in Taiwan.

Empathy plays a crucial role in current education and student development. It is considered as an integral part of the teaching process for fostering social and emotional growth [13]. Research by Smith and Hu [12] suggested that important skills required by 21st-century learners, such as empathy, systems thinking, creativity, numeracy literacy, and abstract reasoning, are challenging to teach. Other studies have highlighted virtual reality as a promising tool to address these educational challenges [6]. As a novel medium, virtual reality offers a unique means to evoke empathy by providing immersive experiences from different perspectives. Previous studies have demonstrated the crucial role of situational empathy in inter-group contact and intervention, as empathy among members of stigmatized groups can improve attitudes toward the target and the entire outgroup [1]. The research by Maunder, White and Verrelli [9] indicated that computer-mediated experiences through virtual reality can reduce fear, anger, and stereotyping of people with schizophrenia.

Empathy is a complex phenomenon with various facets of understanding. While research has demonstrated the brain and body's response to the pain and emotions of others [10], there is still much to explore regarding the role of media in mediating empathy. Virtual reality possesses the properties of immersion and presence. It has the potential to provide meaningful benefits to individuals' cognition and behavior. The perspectives presented within virtual reality environments have the capacity to cultivate empathy by enabling individuals to directly experience virtual representations that aim to enhance specific social objectives, thereby facilitating the adoption of a more empathetic perspective.

Previous studies [4] have found that utilizing a free-exploration approach to experience typical symptoms of mental illnesses within a virtual reality environment resulted in an increase in empathy toward individuals with mental illnesses. In another study, Li, Tang, Tong, Patibanda, Mueller and Liang [8] conducted research and developed two VR games, namely "Myopic Bike" and "Say Hi," to help non-myopic people understand the challenges of myopia and promote empathy for myopic patients. However, previous research has also shown that empathy does not always increase after experiencing virtual reality. For instance, Hadjipanayi and Michael-Grigoriou [5] found that there was no significant improvement in empathy when simulating the symptoms of Asperger's syndrome from a

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first-person perspective, although the VR simulation enhanced participants' understanding of Asperger's syndrome and increased social awareness. In another study by Stelzmann, Toth and Schieferdecker [14], a young individual with schizophrenia was filmed using a 360-degree camera equipped with multiple lenses. Participants in the study were presented from an observer's perspective. The results showed that the virtual reality group exhibited higher levels of stigmatization compared to the control and conventional groups. Although they displayed greater social distancing, they also experienced heightened anxiety and demonstrated slightly less empathy and compassion.

Previous studies have provided meaningful insights into various perspectives on empathy. However, the extent to which virtual reality interventions can enhance empathy towards different groups and under what circumstances remains inconclusive to date. The results have been mixed regarding whether there is a significant improvement in empathy. Additionally, the effective methods for enhancing empathy are still relatively unresolved research questions. Therefore, further exploration and investigation into this subject are warranted. Providing participants with a first-person embodied experience in the virtual reality immersion, they can develop social cognition and form expectations related to their avatars [4]. However, if the design of the simulation adopts a social observation perspective [14], participants may not perceive themselves as embodying a first-person perspective within the virtual environment and may be restricted in their freedom of movement. This limitation has the potential to reinforce stigmatized attitudes, particularly when patients' uncontrollable disease manifestations are represented. Previous research [17] has demonstrated that empathy can be effectively enhanced when virtual reality experiences are designed from a first-person perspective.

2 SYSTEM DESIGN AND IMPLEMENTATION

In the field of neuroscience, Zaki and Ochsner [18] conducted research on empathy, highlighting that the empathy process encompasses various interconnected yet distinct processes in which the "perceiver" becomes attuned to the "goal". Drawing from this understanding, the system experience in this study was built upon the original experiential script, and all participants engaged in the experiment from a first-person perspective. To create a closer resemblance to daily-life situations, this study focuses on the activity of going from home to the supermarket for shopping, incorporating real-life environments as the main design theme. During this simulated experience, the interaction design is added to the experience process, and the positive symptoms of schizophrenia, including hallucinations, delusions, and disordered thinking, are simulated. The objective is to allow participants to gain insights into how seemingly mundane actions in daily life may be influenced by the symptoms of schizophrenia, leading to divergent modes of thinking, emotions, and perceptions that impact cognition, daily behaviors, and thoughts.

In this study, we investigated the potential of utilizing VR exposure as a direct and active approach to understanding the influence of situational empathy, without the introduction of perspective-taking guidance. Through this approach, we consider VR exposure as a means to induce individuals to develop empathy towards others

and bring about changes in attitudes following positive awareness experiences.

The development of the system was done by the Unity3D game engine and it is implemented on the HTC VIVE FOCUS3, enabling the display of immersive 3D scenes. Subsequently, mental health professionals were invited to evaluate the system to verify whether the virtual reality simulation system closely replicates realistic characteristics of thought disorders.

To familiarize the participants with the operation of the experiential system, the initial scene of the experiment will place them in a hall where voice instructions and guidance can assist participants when engaging in short operational exercises. The participants are required to use controllers for movement and interaction within the virtual environment. In order to make them operate more smoothly in the next experience, the participants are prompted to walk around and reach out to touch the ball in front of them initially, allowing them to become familiar with the controller and the sensation of the virtual space.

Once the exercise is completed, the scene will transition to the living room environment, serving as the first scene (Figure 1). At this point, narration will guide the participants through a series of experiential tasks following the plot's development. The living room scene will undergo various changes, such as the presence of psychedelic pictures on the TV and constant transformations of the wall paintings. As the auditory hallucinations persist, participants will experience a sense of being controlled by external voices. Although they have seen the keys before leaving, they will struggle to locate them. The design of VR simulation includes a series of interactive operations within the virtual world to enhance the process of shared experience, enabling participants to embody the target individual and gain insight into the internal state during an episode of the illness.

Upon successful completion of the experiential tasks in the first scene, the scene will move to a street environment. Participants will walk along the street (Figure 2), with a sense of surveillance throughout the process. They will encounter passers-by engaged in conversations that gradually disturb the participants. Over time, they will experience dizziness and a sense of immobility in the surrounding scene. However, they will slowly return to their original appearance once the pedestrians have passed by.

In the third scene (Figure 3), after participants successfully arrive at the entrance of the supermarket, they will feel that the items they intend to purchase have been poisoned with by someone else. When they pick up the fruit, it will appear rotten, but when they put it back down, it will return to the normal appearance.

3 STUDY METHOD

This study adopts a quasi-experimental design to compare changes in empathy and attitudes towards individuals after experiencing the daily life situations of patients with schizophrenia during the onset of positive symptoms through virtual reality, using a questionnaire survey. Before the experiment, the subjects will fill out the Interpersonal Response Inventory (IRI) to account for individual differences in empathy abilities among the research participants and to reduce the influence of personality traits on empathy and attitude changes.

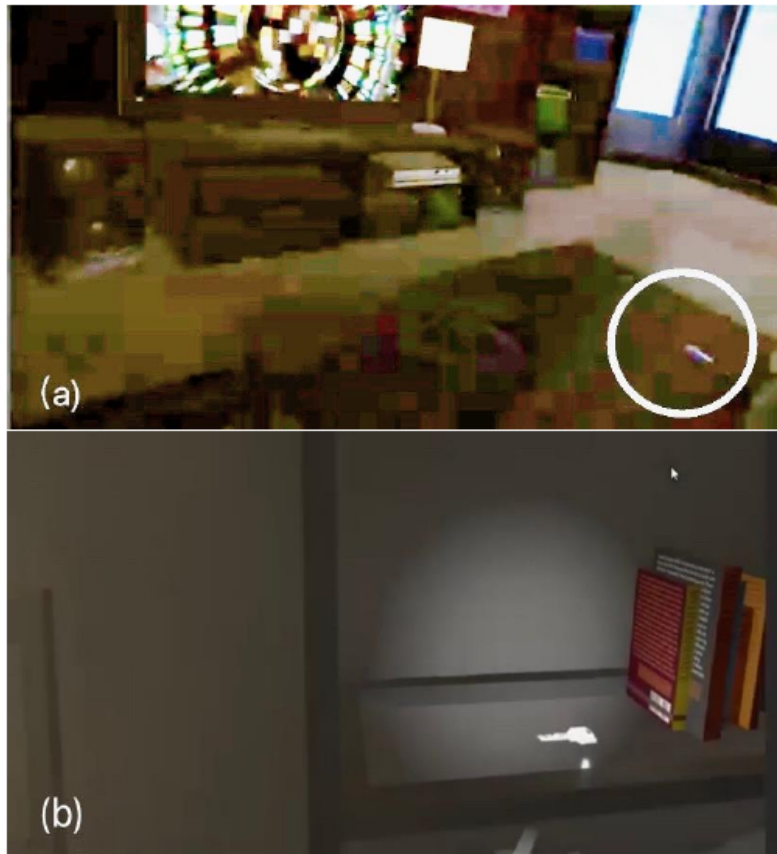


Figure 1: When they believed the key was on the table (a), it was actually an illusion; the real key was on the shelf (b).



Figure 2: Abstract representation of being watched by a multitude of eyes

In this study, the independent variable is virtual reality, the mediating variable is presence, and the dependent variables are empathy toward people with schizophrenia and attitudes toward people with schizophrenia (Figure 4).

3.1 Instruments

The instruments used in the experiment are the "Interpersonal Reactivity Scale", the questionnaire of "Empathy Toward People with Schizophrenia", the questionnaire of "Attitudes Toward People with Schizophrenia", and the questionnaire of "Presence".



Figure 3: When picking up the fruit to be bought and the fruit will turn black

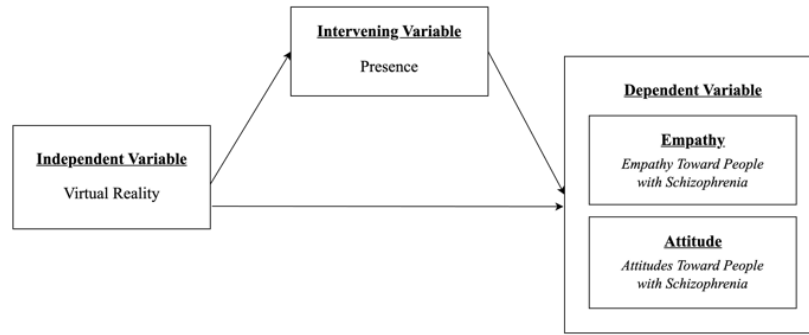


Figure 4: Research architecture and variables

3.1.1 Interpersonal Reactivity Index. The Interpersonal Reactivity Index (IRI), based on empathy traits and revised by Davis [3], is utilized to assess individual levels of empathy. The Interpersonal Responsiveness Scale is a self-report instrument comprising 4 subscales, with participants rating their responses on a 7-point Likert scale. This scale consists of a total of 28 items and is designed to assess individual variations in empathy.

3.1.2 Empathy Toward People with Schizophrenia. To evaluate the participants' empathic feelings towards individuals with schizophrenia, the questionnaire of Empathy Toward People with Schizophrenia was employed. This questionnaire (7-point Likert scale), adapted from Baston [2] by Kalyanaraman et al.[7], is aimed to measure the extent to which participants reported experiencing empathic emotions.

These adjectives can be classified into two categories: personal pain and empathetic emotion. This scale can be utilized either as a difference score, obtained by subtracting individual distress scores from the total score initially proposed by Batson, Fultz and Schoenrade [2], or as a two-dimensional scale with distinct scores for personal distress and empathic interest. In this study, separate

scores on the two subscales will be employed to examine the potential effects of different media exposures on the two components of empathy.

3.1.3 Attitudes Toward People with Schizophrenia. To gauge the participants' attitudes towards schizophrenia, the Attitudes Toward People with Schizophrenia questionnaire, adapted from Baston [2] by Kalyanaraman et al. [7], was administered. This questionnaire (7-point Likert scale) was designed to assess participants' descriptions of their attitudes towards the disorder. Higher scores correspond to more positive attitudes.

3.1.4 Presence Questionnaire. The Presence Questionnaire (PQ), which measures an individual's sense of presence in a virtual environment, was utilized in this study. The revised second version of the PQ, originally developed by Witmer & Singer [15] and further revised by the UQO Cyberpsychology Lab in 2004, was employed. The questionnaire utilized a 7-point Likert scale for participants' responses.



Figure 5: The participants operating the VR system to experience the situations encountered by individuals schizophrenia.

Table 1: samples t-test results of the pilot study

Variables	S.D.	Average	t Value	Significance
Empathy	5.10	4.60	-2.85	.019
Attitude	5.33	3.00	-1.78	.109

3.2 Procedures

The pilot study employed quasi-experimental research, utilizing a pre-post-test approach to examine the impact on emotional and cognitive empathy, as well as attitudinal changes among participants. To facilitate an interactive experience, participants were equipped with a virtual reality head-mounted display device and utilized a controller for navigation and interaction (Figure 5).

4 PILOT STUDY & RESULT

The pilot study recruited 10 participants over the age of twenty years who met the inclusion criteria (5 males, 5 females), ranging from 20 to 46 years old. The experimental results (1) were analyzed using SPSS. In the empathy variables, the average scores were 26.50 on the pre-test and 31.10 on the post-test, $t(9) = -4.60$, $p = 0.019$, a statistically significant relationship. The mean scores of the pre-test and post-test demonstrate a significant difference, with the post-test mean score being higher than the pre-test mean score. This indicates an increase in scores on the empathy dimension following the experience. In the attitudinal variables, the average scores were 39.70 on the pre-test and 42.70 on the post-test, $t(9) = -1.78$, $p = 0.109$, which did not reach statistically significant relationships.

The Pearson Correlation indicates a significant positive correlation of 0.69 ($p = 0.026$) between empathy and presence, demonstrating a large level of association between the two variables. This indicates a positive covariation, implying that they have a positive relationship with each other. And the Pearson Correlation between attitude and presence is 0.508, indicating a moderate level of association. However, the p -value is 0.134 > 0.05 , it can be concluded that there is no significant association between attitude and presence.

5 FUTURE WORK & CONCLUSION

In the early stage of this work, the results from the pilot study indicate that this research has the potential to elicit more positive empathic responses among individuals with psychosis. Compared

with other stimuli, virtual reality can provide a more realistic experience [11], making the stimuli more immersive for individuals. This heightened sense of presence increases the degree of empathy and can lead to a greater empathetic response towards specific targets. However, it is important to note that there was not a significant increase in attitudinal levels. Therefore, it is necessary to gain a deeper understanding of the underlying processes and enhance the effectiveness of the intervention.

Next, we will use the pilot study results of the current study to explore further the correlation between the variables and to improve the experiential system. We will then recruit more people of all ages and genders to explore the potential of virtual reality to change empathy and attitudes for people with psychosis. It is expected that this study emphasizes the use of first-person virtual reality to provide a deeper sense of "imagining the other", as this is the essence of exploring the potential for empathy and attitude change.

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Ethics Statement

Due to the involvement of human participants, this study was reviewed and approved by the Research Ethics Committee of the National Tsing Hua University (REC No.11112HT112).

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